SUMMARY TABLE: GENERAL SCIENCE GRADE 10

Sub-Strand 1.3: Nutrition in Animals — Teacher Reference (Pre-filled)

SUMMARY TABLE: GENERAL SCIENCE GRADE 10	
Sub-Strand	Sub-Strand 1.3: Nutrition in Animals
Driving Question	DRIVING QUESTION: How is the human digestive system adapted to break down and absorb the nutrients in a typical Kenyan meal — and what happens when those adaptations fail?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 **The Ugali and Sukuma Wiki Puzzle — Launching the Anchoring Phenomenon**	Teacher presents the anchoring phenomenon: a plate of ugali, sukuma wiki, and a piece of nyama choma. Learners observe the meal and are asked, 'What happens to this food from the moment it enters your mouth until your body uses it?' Students record initial observations about the meal's physical properties — texture of ugali (starchy, dense), fibrous structure of sukuma wiki, and protein-rich meat. Teacher facilitates a KWL-style elicitation using sticky notes on the Driving Question Board (DQB), sorting learner ideas into 'We think we know,' 'We're not sure,' and 'We wonder.' Key observable tensions surfaced: food is solid but blood is liquid — so how does solid ugali become usable energy?	Learners surface and organise prior knowledge about digestion, distinguishing confident knowledge from uncertain hunches and genuine gaps. They identify that the body must somehow transform food into a usable form, and that this transformation involves multiple body parts. They recognise that different nutrients (starch in ugali, protein in meat, vitamins in sukuma wiki) may be processed differently. The DQB is populated for the first time, establishing the unit's central questions. Teacher note: Do not correct misconceptions at this stage — catalogue them. Common Kenyan learner misconceptions to watch for include: 'food goes directly to blood,' 'the stomach does all the digestion,' and 'ugali gives you energy immediately after eating.'	The anchoring phenomenon (the Kenyan meal) is intentionally chosen because every learner has eaten ugali, sukuma wiki, and nyama choma. This personal relevance lowers affective barriers and activates prior schema. The lesson establishes the unit's explanatory target: the digestive system must be structurally and chemically adapted to transform solid, complex food molecules into small, soluble molecules that can enter the bloodstream. The gap between 'solid ugali' and 'usable glucose in blood cells' is the productive tension that drives all eight lessons. Teacher note: Emphasise to learners that not knowing is the starting point of science — reference Kenyan biochemist Prof. Miriam Were's work on evidence-based health practice as a model of scientific humility.
Lesson 2 **Mechanical vs. Chemical Digestion — Breaking Down the Ugali and Sukuma Wiki**	Teacher demonstrates two parallel processes using tangible analogies: (1) tearing a chapati by hand (mechanical — changes size and shape, not chemical identity) versus (2) adding vinegar to baking soda (chemical — produces a new	Mechanical digestion physically breaks food into smaller pieces — increasing surface area — without changing the chemical nature of the molecules. It is performed by teeth (incisors cut, molars grind ugali and meat), the tongue (shapes the bolus), and	The distinction between mechanical and chemical digestion explains a key adaptation of the digestive system: teeth are mechanically shaped to maximise surface area exposure (incisors for cutting githeri beans, molars for grinding ugali),

	substance). Learners then observe a short video or diagram showing teeth grinding maize-based ugali and stomach churning. A second observation involves iodine solution turning blue-black on a piece of ugali (confirming starch) and remaining amber on a piece of already-digested starch solution — showing that chemical digestion changes the molecule itself. Teacher asks: 'Which process changes WHAT the food is, and which only changes HOW BIG it is?'	stomach muscles (churning). Chemical digestion uses enzymes and acids to break covalent bonds, converting large insoluble molecules into small soluble ones: starch → maltose → glucose; protein → amino acids; fats → fatty acids + glycerol. Both processes are necessary and complementary: mechanical digestion increases surface area so enzymes can work faster. Teacher note: Use the analogy of a panga (machete) cutting a large piece of nyama choma into smaller pieces (mechanical) versus marinating the meat to tenderise it chemically.	while enzyme specificity ensures each nutrient type is broken down by the correct catalyst. This lesson directly addresses the DQB question 'Does the stomach do all the work?' — establishing that digestion is a multi-organ, multi-strategy process. Learners update the DQB, replacing vague ideas with precise distinctions. Teacher note: Cold-call at least 4 learners to explain in their own words why increasing surface area speeds up chemical digestion — wait 10 seconds after posing the question before accepting responses.
Lesson 3 **Enzymes and Regions of the Alimentary Canal — The Body's Chemical Toolkit**	Teacher guides learners through a region-by-region mapping activity using a large labelled diagram of the alimentary canal. For each region, learners record: the enzyme(s) present, the substrate acted upon, and the product formed. Key observations: salivary amylase is active in the mouth (pH ~7) and breaks starch in ugali into maltose; pepsin is active in the stomach (pH ~2) and begins protein digestion of nyama choma; pancreatic amylase, lipase, and proteases act in the small intestine. Learners observe a pH-indicator demonstration showing that stomach acid denatures salivary amylase — establishing that enzyme activity is region- and pH-dependent. Teacher asks: 'Why does the body need different enzymes in different regions, rather than one universal enzyme?'	Enzymes are biological catalysts that are substrate-specific and pH-dependent. The alimentary canal is divided into functional regions, each with an optimised chemical environment: mouth (amylase, pH 7); stomach (pepsin + HCl, pH 1.5–2); small intestine (pancreatic amylase, lipase, peptidases, pH 7–8, aided by bile from the liver which emulsifies fats from nyama choma). Each enzyme has an active site complementary only to its specific substrate — the lock-and-key model. Denaturation (e.g., pepsin denaturing salivary amylase in the stomach) is an irreversible structural change caused by extreme pH or temperature. Teacher note: Emphasise that bile is NOT an enzyme — it is an emulsifier that increases fat surface area for lipase action. This is a high-frequency exam misconception.	The region-specific deployment of enzymes explains a critical adaptation: the digestive system is not a single uniform tube but a series of specialised micro-environments, each optimised for a specific chemical reaction. This directly answers the DQB sub-question 'Why does food take so long to digest?' — because multiple sequential enzyme reactions must occur in specific order and location. The starch in ugali must be partially broken down by salivary amylase in the mouth before pancreatic amylase completes the process in the small intestine; neither enzyme alone is sufficient. Learners update the DQB with the new mechanistic explanation. Teacher note: Ask learners to predict what would happen to protein digestion if a person produced no stomach acid — wait 15 seconds, then cold-call 3 learners before explaining.
Lesson 4 **Lesson 4**	Teacher facilitates a Food Tests Practical (Part 1) focused on detecting starch and reducing sugars in components of the Kenyan meal. Learners test samples of ugali solution (starch-rich), diluted honey (glucose-rich), and sukuma wiki extract using iodine solution and Benedict's reagent respectively. Observations recorded: iodine turns blue-black in the presence of ugali extract (confirming starch); Benedict's reagent turns from blue to brick-red/orange when heated with a glucose-containing	Iodine solution (iodine in potassium iodide) is a test for starch: a blue-black colour indicates starch is present; amber/yellow indicates starch is absent. Benedict's reagent tests for reducing sugars (including glucose and maltose): a colour change from blue → green → yellow → orange → brick-red indicates increasing concentrations of reducing sugar; no colour change (remains blue) indicates no reducing sugar. These chemical tests provide indirect evidence of digestion: if starch disappears and reducing	Food tests are diagnostic tools that reveal the molecular composition of food — they make the invisible (molecules) observable. This lesson provides the evidential foundation for understanding digestion: by comparing test results before and after enzyme treatment, learners can empirically confirm that chemical digestion converts starch (large, insoluble) into reducing sugars (small, soluble). This directly links to the anchoring phenomenon — the ugali on the plate contains starch that tests blue-

	sample (confirming reducing sugars). Learners also test a post-digestion ugali sample (amylase-treated) with iodine — iodine does NOT turn blue-black, confirming starch has been broken down. Teacher asks: 'What does the colour change tell us about what is present — and what does the absence of a colour change tell us?'	sugars appear in the same sample after amylase treatment, digestion has occurred. Teacher note: Ensure learners understand that Benedict's test requires heating in a water bath — cold-calling on safety procedures (e.g., 'What hazard does the water bath present and how do we manage it?') reinforces STEM safety competencies mandated by KICD.	black with iodine, but the glucose absorbed into the bloodstream would not. The transformation between these two states is digestion. Learners update the DQB: 'We now have chemical evidence that starch is broken down into sugars.' Teacher note: Connect to Kenyan agricultural context — farmers at Mwea Irrigation Scheme grow rice (another starch source) and the same iodine test would confirm starch in rice flour.
Lesson 5 **Food Tests Practical (Part 2) — Detecting Proteins and Fats in the Kenyan Meal**	Learners perform Biuret test on a nyama choma (meat) extract and a control (water): meat extract turns violet/purple, confirming protein presence; water remains blue. Learners perform the ethanol emulsion test on a fat-containing sample (cooking oil diluted in ethanol, then water added): a milky-white emulsion forms, confirming fat presence; water control remains clear. Learners also test sukuma wiki extract for protein (mild positive — reflects lower protein content than meat) and compare results across all Kenyan meal components. Teacher facilitates a class results table and asks: 'Which nutrient is most concentrated in nyama choma? Which is most concentrated in ugali? What does this suggest about how different parts of the meal serve different functions in the body?'	The Biuret test detects proteins: Biuret reagent (NaOH + $CuSO_4$) reacts with peptide bonds; violet/purple = protein present; blue = protein absent. The ethanol emulsion test detects fats/lipids: fat dissolves in ethanol; when water is added, the fat re-precipitates as a milky-white emulsion; no emulsion = no fat. Combining all four food tests (iodine, Benedict's, Biuret, ethanol emulsion) allows complete nutritional profiling of any food. Nyama choma is protein- and fat-rich; ugali is starch-rich; sukuma wiki provides vitamins, minerals, and some protein. Each nutrient requires a different digestive strategy — proteins need proteases, fats need lipase and bile, starch needs amylases. Teacher note: Highlight that food tests identify nutrient CLASS, not specific molecules — Biuret detects all proteins, not specifically myosin or actin from meat.	Food tests complete the evidential toolkit learners need to understand WHY the digestive system has different enzymes for different regions: because different nutrients are chemically distinct and require different enzymatic strategies. The Kenyan meal is a nutritionally diverse input — starch (ugali), protein (nyama choma), fat (cooking oil/meat fat), vitamins (sukuma wiki) — and the digestive system's structural and chemical complexity is a direct adaptation to this diversity. This lesson resolves the DQB question 'How does the body know which enzyme to use?' — the answer is structural: each region produces only the enzymes appropriate to its substrate. Learners update the DQB and begin to see the full system. Teacher note: Ask learners to predict what food test result would occur if a person had a condition preventing lipase production — wait 10 seconds, cold-call 3 learners.
Lesson 6 **Adaptations of the Human Digestive System — Structure Meets Function**	Teacher displays a cross-sectional diagram (or physical model) of the small intestine showing villi and microvilli. Learners calculate: if a 6-metre small intestine had no villi, its absorptive surface area would be approximately 0.33 m²; with villi and microvilli, it is approximately 200 m² — roughly the size of a volleyball court. Learners observe diagrams showing the lacteal (for fat absorption), capillary network (for glucose and amino acid absorption), and thin epithelial wall (one cell thick — short diffusion distance). Teacher also shows a diagram of the stomach wall	The small intestine has multiple structural adaptations for maximum absorption: (1) villi — finger-like projections increasing surface area ~10-fold; (2) microvilli (brush border) — further increasing surface area ~20-fold; (3) thin epithelium — one cell thick, minimising diffusion distance; (4) rich capillary network — maintaining concentration gradient for glucose and amino acid absorption; (5) lacteals — absorbing fatty acids and glycerol into the lymphatic system. The stomach is adapted with: rugae (folds allowing expansion to ~1.5L after a large githeri meal); goblet cells	This lesson is the structural climax of the unit: every adaptation described is a direct answer to the driving question. The digestive system is not generically constructed — it is precisely engineered at the microscopic level to maximise the absorption of nutrients from a complex meal. The villi of a Kenyan farmer in Kericho absorbing glucose from ugali and amino acids from beans function identically to those of an athlete in Iten absorbing protein after training — the structure is universal, but its efficiency can be compromised by disease, malnutrition, or

	(rugae for expansion, goblet cells for mucus protecting against self-digestion by HCl). Teacher asks: 'How does each structural feature directly improve the efficiency of digestion or absorption?'	secreting mucus (preventing self-digestion by pepsin and HCl); oxyntic cells secreting HCl (activating pepsinogen → pepsin). When adaptations fail — e.g., coeliac disease destroying villi, or H. pylori bacteria degrading mucus — malabsorption and ulcers result. Teacher note: The 'adaptation failure' component directly addresses the DQB's 'what happens when adaptations fail?' clause — do not skip this.	parasitic infection (e.g., hookworm damaging intestinal lining, prevalent in parts of Kenya). Learners update the DQB with structural evidence. Teacher note: Cold-call 5 learners to match each structural feature to its specific function before moving to model building — allow 15 seconds wait time per question.
Lesson 7 **Model Building — Making a 3D Model of the Human Digestive System**	Learners work in groups to construct a 3D model of the human digestive system using locally available materials (cardboard, string, balloons, clay, maize husks, fabric). Each group must represent: (1) at least 4 organs of the alimentary canal with correct relative sizes and positions; (2) at least 3 enzymes with their region and substrate labelled; (3) at least 2 structural adaptations (e.g., villi, rugae); (4) the pathway of at least one nutrient from the Kenyan meal (e.g., starch from ugali → mouth → small intestine → bloodstream). Groups observe each other's models in a gallery walk, using a structured peer-review protocol: 'One thing that is scientifically accurate,' 'One thing that needs revision,' 'One question I still have.'	Learners integrate knowledge from all six prior lessons into a single coherent explanatory model: digestion is both mechanical (teeth, stomach churning) and chemical (enzymes); enzymes are region-specific and pH-dependent; structural adaptations (villi, microvilli, rugae, mucus) directly serve digestive and absorptive functions; different nutrients from the Kenyan meal follow different absorption pathways (glucose/amino acids → capillaries → portal vein → liver; fatty acids/glycerol → lacteals → lymphatic system). The model-building process surfaces remaining misconceptions and gaps, which are addressed during the gallery walk debrief. Teacher note: Circulate during construction and ask probing questions: 'Where exactly does the starch in this ugali piece first encounter an enzyme?' and 'Show me where the protein from this nyama choma piece gets absorbed.' Allow 5–7 minutes minimum for independent group work before intervening.	Model building is a core NGSS Science and Engineering Practice (Developing and Using Models). The 3D model externalises internal understanding — it makes learners' mental models visible, allowing the teacher to diagnose misconceptions in real time. The act of deciding 'where to put the villi' and 'how to show enzyme specificity' requires deeper processing than passive note-taking. The gallery walk introduces collaborative peer assessment, a KICD-mandated competency. The Kenyan meal remains the anchor: every organ represented should be evaluated by the question 'What does this organ do to the ugali, sukuma wiki, or nyama choma?' Teacher note: Groups that produce incomplete or inaccurate models should not be corrected immediately — use the gallery walk peer questions to guide self-correction. Only intervene if a fundamental misconception (e.g., 'digestion happens in the liver') would mislead other groups.
Lesson 8 **DQB Synthesis & Assessment — Driving Question Resolution and Unit Wrap-Up**	Teacher returns to the original DQB from Lesson 1. Learners re-examine every sticky note posted during the unit — sorting into: 'We now know this and can explain it,' 'We partially understand this,' and 'This is still genuinely unknown to us.' The class constructs a collective final answer to the driving question by tracing the fate of ugali, sukuma wiki, and nyama choma through the entire digestive system — from mechanical breakdown in the mouth to villi absorption in the small intestine to waste	The human digestive system is structurally and chemically adapted at every region to process the specific nutrients in a Kenyan meal: mechanical digestion (teeth, stomach) increases surface area for enzyme action; chemical digestion (salivary amylase, pepsin, pancreatic enzymes, lipase) sequentially breaks macromolecules into absorbable monomers; the small intestine's villi and microvilli maximise absorption surface area (up to 200 m²); different nutrients are absorbed via different	This lesson completes the explanatory arc of the unit. The driving question was intentionally designed to have two clauses — 'how is the system adapted' and 'what happens when adaptations fail' — because science education requires learners to understand both normal function and dysfunction. A Kenyan health context makes this personally relevant: peptic ulcers are common in Kenya (often linked to H. pylori infection and NSAID use); coeliac disease is underdiagnosed; enzyme

	elimination. Teacher also revisits the 'adaptation failure' clause: learners explain what happens when villi are destroyed (coeliac disease — malabsorption of all nutrients), when mucus is insufficient (peptic ulcers — stomach acid erodes lining), and when enzyme production fails (e.g., lactase deficiency — inability to digest milk sugar, relevant to some Kenyan communities).	pathways (capillaries vs. lacteals). When these adaptations fail, specific, predictable consequences follow — malabsorption, ulceration, or nutrient deficiency. Learners can now fully answer the driving question with mechanistic, evidence-based explanations drawing on food test data, structural diagrams, enzyme specificity, and their 3D models. Teacher note: The assessment in this lesson should be formative-to-summative — use the DQB resolution as a structured oral assessment before any written component.	deficiencies affect nutritional status. By connecting digestive system biology to real Kenyan health outcomes, learners see science as a tool for understanding their own bodies and communities. Teacher note: Close the unit by asking learners to write a 'letter to a friend' explaining — in plain Swahili or English — why a person with damaged villi would feel tired and malnourished even if they are eating enough ugali and nyama choma. This transfer task is the highest-order assessment of the unit's core learning.